

# A New Light-Powered Material Makes Hydrogen From Water

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Chemists at Oregon State University have found a new way to make hydrogen from water using light. The work was led by Kyriakos Stylianou of the university's College of Science. His team developed a material called a **photocatalyst** that can produce hydrogen quickly and efficiently.

Hydrogen is used in fuel cells for vehicles. It is also used to make ammonia, refine metals and manufacture plastics.

A catalyst is a substance that speeds up a chemical reaction without being permanently changed. A **photocatalyst** is similar, but light switches it on. After it absorbs light, it reaches a higher energy state and can use that energy to drive reactions faster. The results were published in the Journal of the American Chemical Society. Stylianou said they could give researchers another tool for reducing **greenhouse gases**.

The team studies **metal organic frameworks**, or **MOFs**. These are crystal-like materials full of tiny pores, built from metal ions and organic molecules called linkers. Scientists have already made nearly 100,000 different **MOFs**. This study focused on one called BVR-19. It has an unusual sulfur-to-sulfur bond that briefly breaks in light, creating very reactive sulfur pieces. Stylianou said the material's sulfur-containing building blocks capture light energy and move electrons, so it does not need an extra expensive metal catalyst. It also forms on its own in water at room temperature.

Most hydrogen today is made from **natural gas**, a process that releases carbon dioxide. It costs about \$1.50 per kilogram, while "green" hydrogen costs roughly \$5 per kilogram. Stylianou said the work gives new design rules for making materials that could bring down the cost of **green hydrogen**.

## Word Treasure

**photocatalyst** -- A material that uses light to speed up a chemical reaction.

**metal organic frameworks** -- Crystal-like materials full of tiny pores, built from metal ions and organic molecules.

**MOFs** -- Short for metal organic frameworks.

**greenhouse gases** -- Gases that trap heat in the atmosphere.

**natural gas** -- A fuel that comes from underground and is the usual source of hydrogen today.

**green hydrogen** -- Hydrogen made in cleaner ways.

## Break it down

**Opening:** chemists use light to make hydrogen from water

Uses of hydrogen

How catalysts and photocatalysts work

MOFs and the BVR-19 material

Why current hydrogen production releases carbon dioxide and costs compared

Design rules for cheaper green hydrogen

## Why it matters

Making hydrogen more cheaply without releasing carbon dioxide could give researchers another tool for cutting greenhouse gases.

## Test what you remember

### 1. Who led the Oregon State research?

- ☐ (A) A team from the OSU library
- ☐ (B) Kyriakos Stylianou
- ☐ (C) A group of high school students from Oregon
- ☐ (D) The U.S. government in Washington

### 2. What are MOFs made of?

- ☐ (A) Sulfur gas and nothing else
- ☐ (B) Only salt water and sand
- ☐ (C) Only plastic and rubber
- ☐ (D) Metal ions and organic molecules called linkers

### 3. About how many different MOFs have scientists already made?

- ☐ (A) About 5 million
- ☐ (B) Nearly 100,000
- ☐ (C) About 100
- ☐ (D) Exactly one

### 4. What happens to the sulfur-to-sulfur bond in light?

- ☐ (A) It briefly breaks and creates very reactive sulfur pieces
- ☐ (B) It freezes the nearby water
- ☐ (C) It turns into a solid metal
- ☐ (D) It grows longer and longer forever

### 5. What is the usual way to make most hydrogen today?

- ☐ (A) From plain air in the sky
- ☐ (B) From natural gas, releasing carbon dioxide
- ☐ (C) From fruit juice at factories
- ☐ (D) From wind alone, with no other source

### 6. What did Stylianou say the work provides?

- ☐ (A) A new kind of battery for phones
- ☐ (B) New design rules for materials that could lower green hydrogen's cost
- ☐ (C) A way to make gold from water
- ☐ (D) A cure for all kinds of pollution

Answer key (try the quiz first!): 1: B 2: D 3: B 4: A 5: B 6: B

**1. Kyriakos Stylianou:** Said the work gives a blueprint for materials that could bring down the cost of green hydrogen.